

Exam Code:2024034

BE 2nd Year Examination
December 2023
Electronics and Telecommunication (Section A & B)
3ETRC3 DATA STRUCTURE

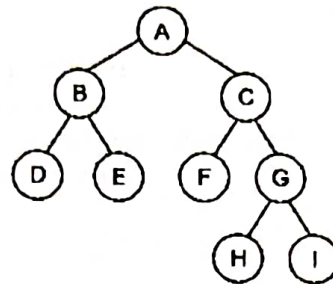
Time: 3 Hrs

Max.Marks: 60

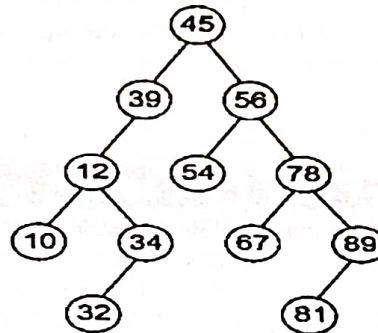
- Note: 1. Attempt any 2 parts of each question, all questions carry equal marks.**
2. Write the output with sample input while answering programming question.
3. Attempt both the parts of question at same place.

- 1 a) Define Data Structures. How are they different from data types? What are different types of data structures?
- b) Consider a two-dimensional array declared as:
`int marks[ ][5]={50,70,50,45,62,84,65,45,74,70,98,45,65,86,38};`
i. Draw the memory map for row major format.
ii. Find out the size of array in Bytes.
iii. Find out the number of rows.
iv. Find out the number of elements in the array.
v. For base address =1000, assuming 4Bytes for "int" data type, find out address of marks[2][3].
- c) Compare Linear Search and Binary Search.
Write C program to find the number = 23 in the following data set using Binary Search algorithm.
`int Data[] = {5,7,9,21,23,35,78,89,93,98};`
- 2 a) Convert the following expression to Reverse Polish notation: (Mention all the steps by using the stack)
Expression: $A - (B / C + (D \% E * F) / G) * H$
- b) Present the algorithms for PUSH() and POP() operation over the STACK(to be implemented as linked list).
- c) Illustrate the quick sort algorithm to sort the following data set.
`int Arr[] = {30,52,29,87,63,27,19,54};`
↑
- 3 a) How Circular queue overcomes limitation of linear queue? Write the algorithms for insertion and deletion in a circular queue. (implemented as array)
- b) Compare and Contrast, Linked List and Arrays. With the help of an example, illustrate creation of singly linked list.
- c) Form a binary max heap and binary min heap for the following data items:
50,40,35,25,20,27,33,58,10
- 4 a) i. Considering the in order and post order traversals, construct the binary tree:
In order: 10 15 20 23 25 30 35 39 42
Post order: 15 10 23 25 20 35 42 39 30

- ii. For the following binary tree, find the in order, pre order and post order traversals

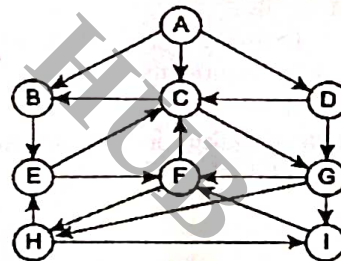


- b) For the Binary Search Tree given below:
 i. Delete the root node
 ii. Insert 11,22,33,44,55,66,77



- c) Create an AVL Tree using the following sequence:
 16, 27, 9, 11, 36, 54, 81, 63, 72

- 5 a) i. Define Graphs. What are applications of graph data structure?
 ii. How are graphs represented? Give the sequential and linked representation of a directed graph.
 b) What is meant by Graph Traversal? Find DFS and BFS for the following graph.



- c) Construct the minimum spanning tree using:
 i. Prim's Algorithm with start Vertex D
 ii. Kruskal's Algorithm

